

COMPUTER SCIENCE DEPARTMENT
MIDTERM EXAMINATION

1. What is the time complexity of binary search in a sorted array of size n ?

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n \log n)$

2. Which data structure operates on a Last-In, First-Out (LIFO) basis?

- A. Queue
- B. Stack
- C. Array
- D. Linked List

3. True or False: A binary tree with n nodes always has height $\log(n)$.

A. True

B. False